

# MT8002 Elements of Set Theory

## 7. Ordered Sets

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Semester 2026-2

# Preliminaries

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## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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7.1 Introduction

7.2 Linearly Ordered Sets

References

# Outline

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## 7.1 Introduction

## 7.2 Linearly Ordered Sets

## References

# Introduction

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- We can use binary relations to order **some** (partial order) or **all** (linear order) the elements of a set.
- Orderings are binary relations that satisfy certain properties.
- Let  $X$  be a set. **Weak** orders  $(\leq_X)$  and **strict** orders  $(<_X)$  can be interchangeably defined.

# Outline

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7.1 Introduction

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# Binary Relations

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## Definition (p. 164)

Let  $X$  be a set. A set  $R$  is a **binary relation on  $X$**  if  $R \subseteq X \times X$ .

## Notation

Let  $R$  be a binary relation on a set  $X$ .

- (a) We write  $(x, y) \in R$  or  $xRy$ .
- (b) We write  $(x, y) \notin R$  or  $x \not R y$ .

# Weak Orders

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## Definitions (p. 164)

Let  $\leq_X$  be a binary relation on a set  $X$ . The relation  $\leq_X$  is a **weak partial order** on the set  $X$  (or the set  $X$  is **weakly partially ordered** by the relation  $\leq_X$ ), if it has the following properties:

- (a) **reflexive**: for all  $x \in X$ ,  $x \leq_X x$ ;
- (b) **anti-symmetric**: for all  $x, y \in X$ , if  $x \leq_X y$  and  $y \leq_X x$  then  $x \doteq y$ ;
- (c) **transitive**: for all  $x, y, z \in X$ , if  $x \leq_X y$  and  $y \leq_X z$  then  $x \leq_X z$ .

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- (c) **transitive**: for all  $x, y, z \in X$ , if  $x \leq_X y$  and  $y \leq_X z$  then  $x \leq_X z$ .

If additionally the relation  $\leq_X$  has the following property:

- (d) **linear**: for all  $x, y \in X$ ,  $x \leq_X y$  or  $y \leq_X x$ ,

then the relation  $\leq_X$  is a **weak linear order** on the set  $X$  (or the set  $X$  is **weakly linearly ordered** by the relation  $\leq_X$ ).

# Weak Orders

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## Examples (informal, p. 164)

The usual relations  $\leq$  and  $\geq$  on the sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  are weak linear orders on those sets.

# Weak Orders

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The usual relations  $\leq$  and  $\geq$  on the sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  are weak linear orders on those sets.

## Example

Let  $X$  be a set. The relation  $\subseteq$  on  $\mathcal{P}(X)$  is a weak partial order.

# Weak Orders

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## Exercise

Let  $X$  be a set. Is the relation  $\subseteq$  on  $\mathcal{P}(X)$  a weak linear order?

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## Exercise

Let  $X$  be a set. Is the relation  $\subseteq$  on  $\mathcal{P}(X)$  a weak linear order?

## Answer

No! For example the relation  $\subseteq$  on the power set of  $\{a, b\}$  is not linear because  $\{a\}, \{b\} \in \mathcal{P}(\{a, b\})$  but neither  $\{a\} \subseteq \{b\}$  nor  $\{b\} \subseteq \{a\}$ .

# Weak Orders

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## Description

The **restriction** of a weak partial order to a subset is the weak partial order obtained by discarding all comparisons involving elements outside of the subset.

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## Definition (p. 164)

Let  $R$  be a weak partial order on a set  $X$  and let  $A \subseteq X$ . The **restriction** of the relation  $R$  to the set  $A$ , denoted  $R|_{A \times A}$ , is the relation defined by

$$R|_{A \times A} := \{ (a, b) \in R : a, b \in A \}.$$

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## Theorem (Exercise 7.2)

Let  $R$  be a weak partial (respectively linear) order on a set  $X$  and let  $A \subseteq X$ . Show that  $R|_{A \times A}$  is a partial (respectively linear) order on the set  $A$ .

# Strict Orders

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## Definitions (p. 165)

Let  $<_X$  be a binary relation on a set  $X$ . The relation  $<_X$  is a **strict partial order** on  $X$  (or the set  $X$  is **strictly partially ordered** by the relation  $<_X$ ) if it has the following properties:

- (a) **irreflexive**: for all  $x \in X$ , it is not the case that  $x <_X x$ ;
- (b) **transitive**: for all  $x, y, z \in X$ , if  $x <_X y$  and  $y <_X z$  then  $x <_X z$ .

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If additionally the relation  $<_X$  has the following property:

- (c) **linear**: for all  $x, y \in X$ ,  $x <_X y$  or  $x \dot{=} y$  or  $y <_X x$ ,
- then the relation  $<_X$  is a **strict linear order** on the set  $X$  (or the set  $X$  is **strictly linearly ordered** by the relation  $\leq_X$ ).

# Strict Orders

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## Examples

The usual relations  $<$  and  $>$  on the sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$  are strict linear orders on those sets.

## Example

Let  $X$  be a set. The relation  $\subset$  on  $\mathcal{P}(X)$  is a strict partial order.

# Strict Orders

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Theorem (relationship between weak and strict partial orders, Exercise 7.4)

(a) Let  $\leq_X$  be a weak partial order on a set  $X$ . The relation

$$x <_X y := x \leq_X y \text{ and } x \neq y$$

is a strict partial order on the set  $X$ .

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Theorem (relationship between weak and strict partial orders, Exercise 7.4)

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$$x <_X y := x \leq_X y \text{ and } x \neq y$$

is a strict partial order on the set  $X$ .

(b) Let  $<_X$  be a strict partial order on a set  $X$ . The relation

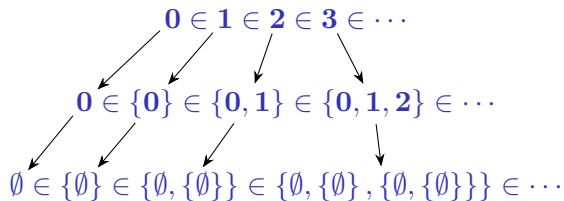
$$x \leq_X y := x <_X y \text{ or } x = y$$

is a weak partial order on the set  $X$ .

# Orders on Natural Numbers

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An “extra” property of natural numbers



# Orders on Natural Numbers

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## Theorem 3.6

The set of natural numbers  $\omega$  is strictly linearly ordered by the relation  $\in$ .

# Orders on Natural Numbers

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## Theorem 3.6

The set of natural numbers  $\omega$  is strictly linearly ordered by the relation  $\in$ .

That is, the relation  $\in$  on  $\omega$  has the following properties:

- (a) **irreflexive**: for all  $n \in \omega$ ,  $n \notin n$ ;
- (b) **transitive**: for all  $m, n, p \in \omega$ , if  $m \in n$  and  $n \in p$  then  $m \in p$ ;
- (c) **linear**: for all  $m, n \in \omega$ ,  $m \in n$  or  $m \doteq n$  or  $n \in m$ .

# Orders on Natural Numbers

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## Example (p. 42)

Although the relation  $\in$  is transitive on the set  $\omega$  it does not mean that the relation is transitive on any set. For example, the relation is not transitive on the set

$$\{0, \{0\}, \{\{0\}\}\}$$

because  $0 \in \{0\}$  and  $\{0\} \in \{\{0\}\}$  but  $0 \notin \{\{0\}\}$ .

# Orders on Natural Numbers

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## Definition (p. 41)

For all  $\mathbf{m}, \mathbf{n} \in \omega$ ,

$$\mathbf{m} < \mathbf{n} := \mathbf{m} \in \mathbf{n},$$

$$\mathbf{m} \leq \mathbf{n} := \mathbf{m} < \mathbf{n} \text{ or } \mathbf{m} \doteq \mathbf{n}.$$

# Conventions

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## Convention

An ordered set  $X$  by a relation  $R$  shall we denote by the pair  $(X, R)$ .

# Order-Isomorphism

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## Description

We need a precise way to express when two ordered sets are “the same”.

# Order-Isomorphism

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We need a precise way to express when two ordered sets are “the same”.

## Definitions (p. 166)

Let  $(X, <_X)$  and  $(Y, <_Y)$  be two strictly partially ordered sets.

The sets are **order-isomorphic**, denoted  $(X, <_X) \cong (Y, <_Y)$ , if there is a **bijection**  $f: X \rightarrow Y$  such that for all  $x_1, x_2 \in X$ ,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

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The map  $f$  is an **order-isomorphism**.

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$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

The map  $f$  is an **order-isomorphism**.

## Remark

The above definition on weakly partially ordered sets is similar.

# Order-Isomorphism

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## Exercise (informal)

Let  $<$  and  $>$  be the usual orders on  $\mathbb{Z}$ . Show that  $(\mathbb{Z}, <) \cong (\mathbb{Z}, >)$ .

# Order-Isomorphism

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## Exercise (informal)

Let  $<$  and  $>$  be the usual orders on  $\mathbb{Z}$ . Show that  $(\mathbb{Z}, <) \cong (\mathbb{Z}, >)$ .

## Solution

Let  $f: \mathbb{Z} \rightarrow \mathbb{Z}$  be the function  $n \mapsto -n$ . We can show that the function  $f$  is a bijection and that it preserves the order, that is, for all  $m, n \in \mathbb{Z}$ ,

$$m < n \quad \text{if and only if} \quad -m > -n.$$

# Order-Isomorphism

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## Theorem (Exercise 7.6)

The relation  $\cong$  is an equivalence relation on strictly partially ordered sets, i.e., the relation  $\cong$  satisfies the following properties:

- (a) **reflexive**: for all  $X$ ,  $X \cong X$ ;
- (b) **symmetric**: for all  $X$  and  $Y$ , if  $X \cong Y$  then  $Y \cong X$ ;
- (c) **transitive**: for all  $X$ ,  $Y$  and  $Z$ , if  $X \cong Y$  and  $Y \cong Z$  then  $X \cong Z$ .

# Order-Isomorphism

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## Definition (p. 168)

An **order-theoretic invariant** is a property of an ordered set preserved under order-isomorphism.

# Order-Embedding

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We also need to define when an ordered set fits inside another while preserving its order structure entirely.

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## Definition (p. 167)

Let  $(X, <_X)$  and  $(Y, <_Y)$  be two strictly partially ordered sets.

A function  $f: X \longrightarrow Y$  is an **order-embedding** of  $(X, <_X)$  into  $(Y, <_Y)$  if  $f$  is **one-one** and, for all  $x_1, x_2 \in X$ ,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

# Order-Embedding

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## Exercise (informal)

Let  $<$  be the usual order on  $\mathbb{N}$  and  $\mathbb{R}$ . Show an order-embedding of  $(\mathbb{N}, <)$  into  $(\mathbb{R}, <)$

# Order-Embedding

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## Exercise (informal)

Let  $<$  be the usual order on  $\mathbb{N}$  and  $\mathbb{R}$ . Show an order-embedding of  $(\mathbb{N}, <)$  into  $(\mathbb{R}, <)$

## Solution

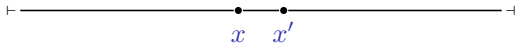
Let  $f: \mathbb{N} \longrightarrow \mathbb{R}$  be the function  $n \longmapsto n$ . We can show that the function  $f$  is one-one and that it preserves the order, that is, for all  $m, n \in \mathbb{N}$ ,

$$m > n \quad \text{if and only if} \quad f(m) > f(n).$$

# Representation of Linearly Ordered Sets

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*“If  $<_x$  is the order on  $X$ , and the elements  $x, x'$  of  $X$  are related by  $x <_x x'$ , we picture  $x$  to the **left** of  $x'$ . [...] Of course, we must not read too much into the picture. The picture is very suggestive of the usual real line, but for instance there might not be any elements of  $X$  between  $x$  and  $x'$  in the picture above; and indeed there might be no order-embedding from  $X$  into  $\mathbb{R}$  with its usual order.” (p. 167)*



# Maximum and Minimum Elements

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## Definitions (p. 168)

Let  $X$  be a set weakly linearly ordered by  $\leq_X$  and let  $a, b \in X$ .

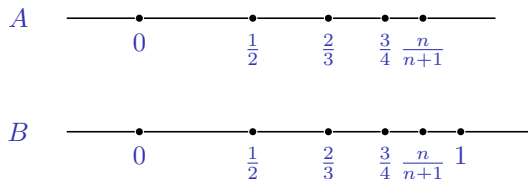
- (a) The element  $b$  is the **maximum** element of  $X$ , if  $x \leq_X b$  for all  $x \in X$ .
- (b) The element  $a$  is the **minimum** element of  $X$ , if  $a \leq_X x$  for all  $x \in X$ .

# Maximum and Minimum Elements

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Example (informal, p. 168)

Let  $A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\}$  and  $B = A \cup \{1\}$ , subsets of  $\mathbb{Q}$ , with the usual order.



The set  $A$  has minimum element  $0$ , but no maximum element.

The set  $B$  has minimum element  $0$ , and maximum element,  $1$ .

# Maximum and Minimum Elements

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## Remark

In the next theorem we shall see that the maximum and the minimum are order-theoretic invariants.

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## Theorem 7.1

Let  $f: X \longrightarrow Y$  be an order-isomorphism from  $(X, \leq_X)$  to  $(Y, \leq_Y)$ .

- (a) If  $X$  has a maximum element then so does  $Y$ .
- (b) If  $X$  has a minimum element then so does  $Y$ .

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In the next theorem we shall see that the maximum and the minimum are order-theoretic invariants.

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- (a) If  $X$  has a maximum element then so does  $Y$ .
- (b) If  $X$  has a minimum element then so does  $Y$ .

## Proof

Suppose that  $X$  has a maximum element  $b$ . Then  $x \leq_X b$  for all  $x \in X$ , so  $f(x) \leq_Y f(b)$  for all  $x \in X$ , as  $f$  is an order-isomorphism. As  $f$  is a bijection, every element of  $Y$  is of the form  $f(x)$  for some  $x \in X$ . Thus  $f(b)$  is the maximum element of  $Y$ . The argument for a minimum element is similar.

# Maximum and Minimum Elements

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## Example (informal)

Let  $A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\}$  and  $B = A \cup \{1\}$ , subsets of  $\mathbb{Q}$ , with the usual order.

From Theorem 7.1 the ordered sets  $(A, <)$  and  $(B, <)$  are not order-isomorphic because the set  $B$  has maximum element, but the set  $A$  does not.

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Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).